Hot 100 80s Charts Application Run Book

This run book explains the requirements, behavior, operating steps, and support notes for the Streamlit application built around the 1980s Billboard Hot 100 dataset.
Application entry point: app.py
Database: hot100-80s.db
Source data: hot-100-80s.csv

Application Requirements

- Load hot-100-80s.csv into a SQLite database named hot100-80s.db.
- Provide a Streamlit app with a main page, year page, song page, and common navigation menu.
- Set the Streamlit page title to ’Welcome to the 80s Hot 100 Charts’.
- On the main page, show the top 10 most successful songs of all time using a score based on Hot 100 position and time on chart.
- On the year page, allow the user to select a chart year. The default is the highest year
available in the database.
- On the year page, show the top 50 songs for the selected year and allow each song to be opened on the song page.
- On the song page, show a chronological Hot 100 position timeline for the selected song.
- Provide a dark mode option and a shared 80s music montage background across all pages.

What The Application Does

- The app presents an interactive view of Hot 100 chart activity from 1980 through 1989. It
ranks songs by a success score that rewards both high chart placement and longevity.
- The success score is calculated as the sum of 101 minus the Hot 100 position for every weekly chart appearance. A number 1 week therefore contributes 100 points, while a number 100 week contributes 1 point.
- The main page visualizes the top 10 songs across the full decade.
- The year page filters rankings to a selected year and supports click-through song navigation.
- The song page shows the selected song’s week-by-week chart history with the y-axis reversed so stronger chart positions appear higher on the chart.
Data And Database

- The CSV file contains one row per song per chart week with these fields: chart_week, current_week, title, performer, last_week, peak_pos, and wks_on_chart.
- load_hot100.py rebuilds the hot100 table from the CSV and derives the year column from chart_week.
- The generated database contains indexes for year, song lookup, and chart week lookup to keep the Streamlit reports responsive.
- The current loaded database has 52,200 chart rows covering years 1980 through 1989.

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How To Run

- Install dependencies with: pip install -r requirements.txt
- Rebuild the database when the CSV changes with: python3 load_hot100.py
- Start the app with: streamlit run app.py
- If streamlit is not on PATH, use: python3 -m streamlit run app.py
- Open the local URL printed by Streamlit, commonly http://localhost:8501 or http://127.0.0.1:8501.

Navigation And User Workflow

- Use the sidebar navigation menu to move between Main Page, Year Page, and Song Page.
- Use the sidebar Dark mode toggle to switch between light and dark visual themes. The background image remains visible on all pages with an overlay that preserves readability.
- On Year Page, select a year from the dropdown. The highest available year is selected by default.
- Click a song button in the top 50 report to open that song on the Song Page.
- On Song Page, the dropdown can also be used to select any song directly.

File Inventory

- app.py: Streamlit application, navigation, queries, charts, theme, and background styling.
- load_hot100.py: CSV-to-SQLite loading script.
- hot-100-80s.csv: Source chart data.
- hot100-80s.db: SQLite database used by the app.
- assets/hot100_80s_montage.png: Shared page background image.
- generate_80s_background.py: Regenerates the background asset.
- requirements.txt: Python package dependencies.
- docs/Hot_100_80s_Run_Book.pdf: This run book.
Validation Checklist

- Run python3 load_hot100.py and confirm it reports 52,200 rows.
- Run python3 -m py_compile app.py load_hot100.py generate_80s_background.py to check syntax.
- Start Streamlit and confirm the app loads without errors.
- Confirm the main page shows a top 10 chart.
- Confirm the year dropdown defaults to 1989.
- Confirm clicking a year-page song opens that song on the song page.
- Confirm the dark mode toggle changes page colors while preserving legibility.

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Troubleshooting

- If the app reports that hot100-80s.db is missing, run python3 load_hot100.py from the project directory.
- If Streamlit is not found, install requirements or run it as a Python module with python3 -m streamlit run app.py.
- If the background image is missing, run python3 generate_80s_background.py.
- If charts are blank, verify that hot100-80s.db contains the hot100 table and that the CSV
load completed successfully.

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